ELDERLY CARE MONITORING SYSTEM WITH EMERGENCY ALERTS

A SRS REPORT

Submitted in the partial fulfilment for the Course of

CSA1012 SOFTWARE ENGINEERING

to the award of the degree of

BACHELOR OF ENGINEERING

IN

INFORMATION TECHNOLOGY

Submitted by

MAHIMA (192521280)

Under the Supervision of

Dr. K. Anita Davamani

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

November 2025

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

Engineer to Excel

DECLARATION

I, **MAHIMA (192521280),** of the **INFORMATION TECHNOLOGY,** Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled **'ELDERLY CARE MONITORING SYSTEM WITH EMERGENCY ALERTS** 'is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place: Chennai

Date: 28-11-2025

Signature of the Students with Names

MAHIMA (192521280)

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "**ELDERLY CARE MONITORING SYSTEM WITH EMERGENCY ALERTS**" has been carried out by **Mahima (192521280)** under the supervision of **Dr. Rameswari** and is submitted in partial fulfilment of the requirements for the current semester of the B. Tech **INFORMATION TECHNOLOGY** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

SIGNATURE

Program Director
Department of IT
Saveetha School of Engineering
SIMATS

SIGNATURE

Professor
Department of IT
Saveetha School of Engineering
SIMATS

Submitted for the Project work Viva-Voce held on ___________________________________
.

INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

We would like to express our heartfelt gratitude to all those who supported and guided us throughout the successful completion of our Capstone Project. We are deeply thankful to our respected Founder and Chancellor, Dr. **N.M. Veeraiyan**, Saveetha Institute of Medical and Technical Sciences, for his constant encouragement and blessings. We also express our sincere thanks to our Pro-Chancellor, **Dr. Deepak Nallaswamy Veeraiyan**, and our Vice-Chancellor, **Dr. S. Suresh Kumar**, for their visionary leadership and moral support during this project.

We are truly grateful to our Director, **Dr. Ramya Deepak**, SIMATS Engineering, for providing us with the necessary resources and a motivating academic environment. Our special thanks to our Principal, **Dr. B. Ramesh** for granting us access to the institute's facilities and encouraging us throughout the process. We sincerely thank our Head of the Department, **Dr K. Malathi** for her continuous support, valuable guidance, and constant motivation.

We are especially indebted to our guides, **Dr. K. Anita Davamani** or their creative suggestions, consistent feedback, and unwavering support during each stage of the project. We also express our gratitude to the Project Coordinators, Review Panel Members (Internal and External), and the entire faculty team for their constructive feedback and valuable inputs that helped improve the quality of our work. Finally, we thank all faculty members, lab technicians, our parents, and friends for their continuous encouragement and support.

Signature With Student Name

MAHIMA (192521280)

INDEX

1. Scenario-Based Requirement Analysis

1.1 Business Domain:

The proposed system belongs to the **Healthcare Technology (HealthTech)** domain, specifically focusing on **elderly healthcare monitoring** using the Internet of Things (IoT), wearable devices, cloud computing, and Artificial Intelligence (AI). The healthcare industry is experiencing a rapid increase in the elderly population, leading to a greater demand for remote patient monitoring and timely medical assistance.

Many senior citizens prefer living independently instead of staying in hospitals or assisted living facilities. However, living alone increases the risk of delayed medical attention during emergencies such as falls, abnormal heart rate, sudden illness, or missed medication. Traditional caregiving methods rely heavily on manual supervision and periodic health check-ups, making continuous monitoring difficult.

The Elderly Care Monitoring System aims to bridge this gap by providing a smart software platform that collects health information from wearable sensors, analyses the data in real time, detects emergencies automatically, and immediately alerts caregivers, family members, or healthcare providers. The system also supports medication reminders, location tracking, and centralized health record management to improve the quality of elderly healthcare services.

1.2 Scenario Understanding:

The healthcare sector is facing new challenges due to an increasing number of elderly individuals who require continuous medical observation while maintaining their independence. Many families cannot provide round-the-clock supervision because of work commitments or geographical distance. As a result, emergencies such as falls, sudden health deterioration, or missed medications may go unnoticed until it is too late.

Current healthcare practices mainly depend on manual caregiving, scheduled hospital visits, and basic wearable devices that provide limited monitoring capabilities. These methods often fail to detect abnormal situations immediately or provide instant communication between elderly users, caregivers, and healthcare professionals.

The proposed Elderly Care Monitoring System addresses these problems by integrating wearable IoT sensors, cloud computing, mobile applications, GPS technology, and intelligent alert mechanisms into a single platform. The system continuously monitors vital signs such as heart rate, body temperature, oxygen saturation, and movement. Whenever abnormal conditions or falls are detected, emergency notifications are automatically sent to registered caregivers and hospitals along with the user's current location. The system also reminds users to take medicines on time and securely stores health records for future medical reference.

This solution improves elderly safety, reduces healthcare risks, enhances communication between stakeholders, and supports independent living.

1.3 Existing Challenges:

The existing elderly care system faces several practical and technical challenges that reduce the effectiveness of healthcare monitoring.

Challenge 1: Lack of Continuous Health Monitoring

Most elderly individuals are monitored only during hospital visits or occasional family interactions. Continuous observation of vital signs is generally unavailable, making it difficult to identify health issues at an early stage.

Challenge 2: Delayed Emergency Response

If an elderly person experiences a fall, heart attack, or sudden illness while alone, family members may not receive immediate information. Delayed emergency response can significantly increase health risks.

Challenge 3: Medication non-compliance

Many elderly individuals forget to take medicines on time or accidentally miss prescribed doses, which may worsen chronic health conditions.

Challenge 4: Difficulty in Remote Monitoring

Family members and caregivers often live in different locations and cannot physically monitor elderly individuals throughout the day. This creates uncertainty regarding their safety and well-being.

Challenge 5: Manual Health Record Management:

Health records are often maintained manually or across different healthcare systems, making it difficult for doctors to access complete medical histories during emergencies.

Challenge 6: Lack of Centralized Monitoring

Caregivers responsible for multiple elderly individuals do not have a centralized platform to monitor all patients simultaneously, resulting in inefficient healthcare management.

Challenge 7: Data Security and Privacy

Medical information is highly sensitive. Existing systems may not provide adequate security measures such as encryption, secure authentication, and controlled access, increasing the risk of unauthorized access and data breaches.

1.4 Stakeholder Analysis:

The successful implementation of the Elderly Care Monitoring System depends on the collaboration of multiple stakeholders. Each stakeholder has specific roles and responsibilities within the system.

Stakeholder	Role	Responsibilities
Elderly User	Primary User	Wears IoT devices, receives medication reminders, views health status, and triggers SOS during emergencies.
Family Members	Care Support	Monitor elderly health remotely, receive emergency alerts, communicate with healthcare providers, and provide immediate assistance when required.
Caregivers	Patient Monitoring	Continuously monitor health data, respond to emergency notifications, update patient information, and ensure daily well-being.
Doctors	Medical Professional	Review patient health reports, analyse medical history, prescribe treatments, and provide healthcare recommendations.
Hospitals / Emergency Medical Services	Emergency Response	Receive emergency alerts, dispatch ambulances, and provide immediate medical treatment during critical situations.
System Administrator	System Management	Manage user accounts, configure IoT devices, maintain cloud infrastructure, ensure system security, and perform software maintenance.
IoT Device Manufacturers	Hardware Provider	Develop wearable sensors capable of collecting accurate health information and transmitting data reliably.

Cloud Service Provider	Infrastructure Provider	Store medical records securely, maintain server availability, perform data backup, and ensure system scalability.

1.5 System Objectives

The proposed Elderly Care Monitoring System has been designed to achieve the following objectives:

- Develop a real-time healthcare monitoring system for elderly individuals.
- Continuously monitor vital parameters including heart rate, body temperature, oxygen saturation, and physical activity.
- Detect falls and abnormal health conditions automatically using wearable sensors.
- Generate instant emergency alerts for caregivers, family members, hospitals, and emergency services.
- Provide medication reminders to improve medicine adherence.
- Enable GPS-based location tracking during emergency situations.
- Offer a centralized dashboard for caregivers to monitor multiple elderly users simultaneously.
- Store medical records securely using cloud-based infrastructure.
- Protect sensitive healthcare information through secure authentication and encryption mechanisms.
- Improve emergency response time and overall patient safety.
- Promote independent living while ensuring continuous healthcare support.

2. Functional Requirements

The following functional requirements describe the core functionalities that the proposed Elderly Care Monitoring System must provide.

Requirement ID	Functional Requirement
FR-01	The system shall allow elderly users, caregivers, doctors, and administrators to register and log in securely.
FR-02	The system shall collect health data from wearable IoT sensors in real time.
FR-03	The system shall continuously monitor heart rate, body temperature, oxygen saturation, and movement.
FR-04	The system shall detect falls using accelerometer and motion sensor data.

FR-05	The system shall identify abnormal health conditions based on predefined threshold values.
FR-06	The system shall automatically generate emergency alerts when abnormal conditions are detected.
FR-07	The system shall send emergency notifications to caregivers, family members, and healthcare providers.
FR-08	The system shall provide GPS location information during emergencies.
FR-09	The system shall allow elderly users to trigger an SOS alert manually.
FR-10	The system shall provide medication reminders according to the prescribed schedule.
FR-11	The system shall notify caregivers if medication is missed.
FR-12	The system shall store patient medical history securely in the cloud.
FR-13	The system shall allow doctors to review patient health reports and medical history.
FR-14	The system shall generate daily, weekly, and monthly health reports.
FR-15	The system shall allow caregivers to monitor multiple elderly users through a centralized dashboard.
FR-16	The system shall maintain a history of emergency alerts and notifications.
FR-17	The system shall provide role-based access control for different users.
FR-18	The system shall maintain activity logs for system monitoring and auditing.
FR-19	The system shall synchronize health data between wearable devices and the cloud server.
FR-20	The system shall notify users when wearable devices lose connectivity or have low battery levels.

3. Non-Functional Requirements

The quality attributes of the system are defined through the following non-functional requirements.

Requirement Category	**Requirement**
Performance	The system should process incoming health data and generate emergency alerts within **5 seconds** of detecting an abnormal condition.
Reliability	The system should provide continuous monitoring with minimal downtime and accurate sensor data processing.
Availability	The system should remain operational **24×7** with an availability target of at least **99.9%**.
Security	All medical information must be encrypted during transmission and storage. Role-based authentication shall prevent unauthorized access.
Privacy	Patient health information shall only be accessible to authorized users in compliance with healthcare privacy regulations.

Usability	The user interface should be simple, intuitive, and designed with large buttons and readable text suitable for elderly users.
Scalability	The system should support the addition of new users, wearable devices, and healthcare providers without affecting performance.
Maintainability	The software shall follow a modular architecture to simplify debugging, updates, and future enhancements.
Portability	The application should operate on Android, iOS, and modern web browsers.
Interoperability	The system should support communication with wearable IoT devices, cloud databases, GPS services, and notification platforms.
Backup and Recovery	Automatic backups shall be performed periodically to prevent loss of medical records during system failures.
Safety	The system shall prioritize emergency alerts and ensure that critical notifications are delivered reliably to caregivers and healthcare providers.

4. Actor Identification and Use Case Modelling

4.1 Primary Actors:

Primary actors directly interact with the Elderly Care Monitoring System to perform day-to-day operations.

Actor	Description	Responsibilities
Elderly User	Senior citizen using the wearable device and mobile application.	View health status, receive medication reminders, send SOS alerts, and monitor personal health information.
Caregiver	Individual responsible for monitoring and assisting elderly users.	Monitor health data, receive emergency alerts, manage medication schedules, and respond to emergencies.
Doctor	Healthcare professional responsible for medical supervision.	Access health reports, review patient history, update prescriptions, and provide treatment recommendations.
System Administrator	Person responsible for managing the entire software system.	Manage users, maintain IoT devices, configure system settings, monitor security, and generate reports.

4.2 Secondary Actors:

Secondary actors support the functioning of the system but do not use it as frequently as primary actors.

Actor	Description	Responsibilities

Family Member	Relative of the elderly user.	Receive emergency notifications, monitor health status remotely, and communicate with caregivers.
Hospital/Emergency Medical Service	Healthcare organization providing emergency medical support.	Receive emergency alerts and dispatch ambulance services.
IoT Wearable Device	Smart wearable sensor used by the elderly.	Collect and transmit vital signs such as heart rate, temperature, oxygen level, and movement.
GPS Service	External location service.	Provide real-time location during emergencies.
Cloud Server	Cloud-based storage platform.	Store health records, synchronize sensor data, and manage backups securely.
Notification Service (SMS/Email/App)	Third-party communication service.	Deliver emergency alerts, medication reminders, and notifications to users.

4.3 Major Use Cases:

Use Case ID	**Use Case**	**Primary Actor**
UC-01	Register/Login	Elderly User, Caregiver, Doctor, Administrator
UC-02	View Personal Dashboard	Elderly User
UC-03	Monitor Vital Signs	IoT Device
UC-04	Detect Fall	IoT Device
UC-05	Generate Emergency Alert	System
UC-06	Send Emergency Notification	System
UC-07	View Health Reports	Caregiver, Doctor
UC-08	Manage Medication Schedule	Caregiver, Doctor
UC-09	Receive Medication Reminder	Elderly User
UC-10	Track Live Location	Caregiver
UC-11	Update Medical Records	Doctor
UC-12	Manage User Accounts	Administrator
UC-13	Manage IoT Devices	Administrator
UC-14	Generate System Reports	Administrator
UC-15	Trigger SOS Alert	Elderly User

4.4 Use Case Descriptions:

Use Case UC-01: Register/Login:

Item	Description
Primary Actor	Elderly User, Caregiver, Doctor, Administrator
Purpose	Authenticate users before accessing the system.
Pre-condition	User must have a registered account.
Main Flow	User enters username and password. The system validates the credentials and grants access based on the assigned role.
Post-condition	User successfully logs into the system dashboard.

Use Case UC-02: Monitor Vital Signs:

Item	Description
Primary Actor	IoT Wearable Device
Purpose	Continuously collect health information from wearable sensors.
Pre-condition	Wearable device must be connected to the system.
Main Flow	Sensors collect heart rate, body temperature, oxygen saturation, and movement data. The information is transmitted to the cloud server.
Post-condition	Health information is updated in the database.

Use Case UC-03: Detect Fall

Item	Description
Primary Actor	IoT Wearable Device
Purpose	Identify accidental falls using motion sensors.
Pre-condition	Sensors are active and connected.
Main Flow	Accelerometer and gyroscope detect sudden movement followed by inactivity. The system confirms the fall event.
Post-condition	Emergency alert is generated.

Use Case UC-04: Generate Emergency Alert

Item	Description
Primary Actor	System
Purpose	Notify concerned individuals during emergencies.
Pre-condition	Abnormal health condition or fall detected.
Main Flow	The system analyses sensor data, generates an emergency alert, and shares the user's location with caregivers and hospitals.
Post-condition	Emergency notification is successfully delivered.

Use Case UC-05: Medication Reminder

Item	Description
Primary Actor	Elderly User
Purpose	Remind users to take medicines on time.
Pre-condition	Medication schedule has been configured.
Main Flow	The system sends reminders through the mobile application or wearable device. The user confirms medication intake.
Post-condition	Medication history is updated.

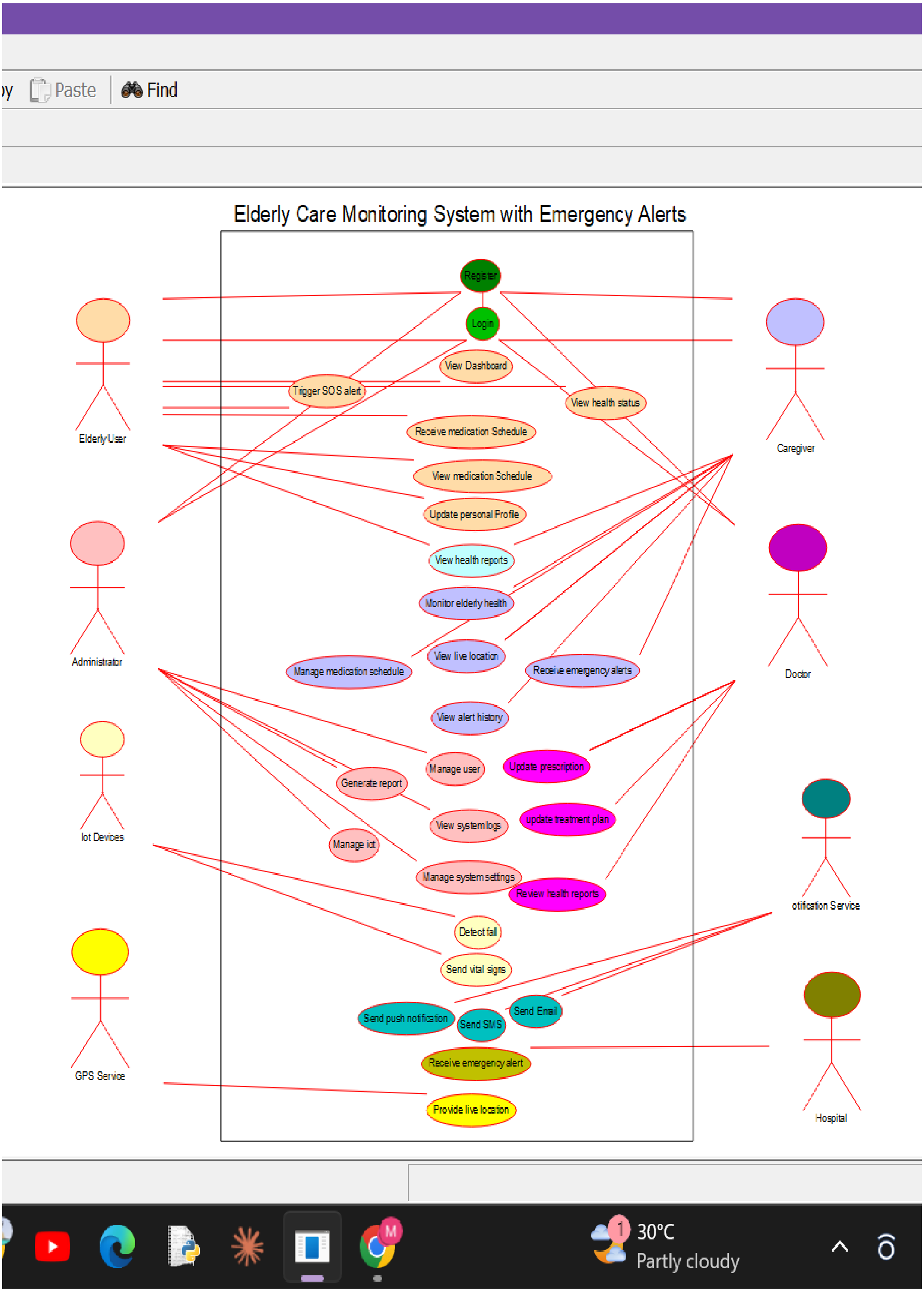
Use Case UC-06: View Health Reports

Item	Description
Primary Actor	Doctor, Caregiver
Purpose	Review the elderly user's health status and history.
Pre-condition	Health data must be available in the database.
Main Flow	User selects a patient and views health trends, reports, and emergency history.
Post-condition	Latest reports are displayed for analysis.

Use Case UC-07: Manage Users

Item	Description
Primary Actor	Administrator
Purpose	Manage user accounts and permissions.
Pre-condition	Administrator is authenticated.
Main Flow	Add, edit, delete, or deactivate users and assign appropriate roles.
Post-condition	User database is updated successfully.

4.5: USE CASE DIAGRAM



Paste
Find
Elderly Care Monitoring System with Emergency Alerts
Register
Login
View Dashboard
Trigger SOS alert
View health status
Receive medication Schedule
View medication Schedule
Update personal Profile
View health reports
Monitor elderly health
View live location
Manage medication schedule
Receive emergency alerts
View alert history
Manage user
Update prescription
Generate report
View system logs
update treatment plan
Manage iot
Manage system settings
Review health reports
Detect fall
Send vital signs
Send push notification
Send SMS
Send Email
Receive emergency alert
Provide live location
Elderly User
Administrator
Iot Devices
GPS Service
Caregiver
Doctor
otification Service
Hospital
30°C
Partly cloudy

5. Software Requirement Specification (SRS)

5.1 Introduction:

The **Elderly Care Monitoring System with Emergency Alerts** is a smart healthcare application developed to improve the safety, health, and independence of elderly individuals. The system uses wearable IoT devices, cloud computing, GPS technology, and intelligent software to continuously monitor health conditions and provide immediate assistance during emergencies.

The purpose of this Software Requirement Specification (SRS) is to define the functional and non-functional requirements of the proposed system. It serves as a communication document between stakeholders, developers, testers, and project managers, ensuring that all system requirements are clearly understood before development begins.

5.2 Problem Description

As the elderly population continues to grow, ensuring their safety has become a major challenge. Many senior citizens live independently without continuous supervision from family members or healthcare professionals. Existing monitoring methods depend largely on manual observation, periodic hospital visits, or basic wearable devices that provide limited functionality.

The absence of continuous monitoring can lead to delayed responses during emergencies such as falls, abnormal heart rate, missed medications, or sudden illness. This delay may result in severe health complications or even loss of life.

The proposed Elderly Care Monitoring System addresses these challenges by providing continuous health monitoring, automatic fall detection, medication reminders, GPS location tracking, secure cloud-based health records, and instant emergency notifications to caregivers and healthcare providers.

5.3 Scope of the System

The system provides a complete digital platform for monitoring and managing elderly healthcare. It supports real-time health monitoring through wearable sensors, automatic emergency detection, secure health record management, medication reminders, and remote caregiver monitoring.

The software allows caregivers and doctors to monitor multiple elderly users simultaneously using a centralized dashboard. It also ensures secure storage of medical information through cloud-based services and provides role-based access control to maintain patient privacy.

The scope of the project includes:

- Continuous monitoring of vital signs.
- Fall detection using wearable sensors.
- Emergency alert generation.
- GPS-based location tracking.
- Medication reminder management.
- Secure cloud-based medical record storage.
- Remote monitoring dashboard.
- Health report generation.
- User authentication and role management.

5.4 Overall Description

The Elderly Care Monitoring System is a cloud-based healthcare solution consisting of wearable IoT devices, a mobile application, a web dashboard, and a centralized cloud server.

Wearable sensors continuously collect health information and transmit it to the cloud. The system analyses the incoming data and detects abnormal conditions such as falls or unusual vital signs. If an emergency occurs, notifications are immediately sent to caregivers, family members, and healthcare providers along with the user's GPS location.

The platform also allows doctors to review health reports, update prescriptions, and maintain medical histories, while administrators manage users, devices, and system security.

5.5 Product Perspective

The proposed system operates as an integrated healthcare platform that connects wearable devices, mobile applications, cloud databases, notification services, and healthcare providers.

Major components include:

- Wearable IoT sensors
- Mobile application
- Web-based caregiver dashboard
- Cloud database
- GPS services
- Notification services (SMS, Email, Mobile Push Notifications)
- Hospital emergency support system

5.6 Data Requirements

The system stores the following data securely in a cloud database:

- User Data: User ID, Name, Contact, Role, Login Credentials.
- Health Data: Heart Rate, Temperature, SpO_2, Blood Pressure, Activity Status.
- Medication Data: Medicine Name, Dosage, Schedule, Reminder Status.
- Emergency Data: Alert ID, Alert Type, Time, GPS Location, Status.
- Medical Records: Health History, Prescriptions, Doctor Notes.

5.7 External Interface Requirements

→ User Interface
  - Mobile application for elderly users.
  - Web dashboard for caregivers, doctors, and administrators.
  - Simple and user-friendly design.

→ Hardware Interface
  - Smartwatch/Fitness Band
  - Heart Rate Sensor
  - Temperature Sensor
  - GPS Module
  - Accelerometer

→ Software Interface
  - Cloud Database
  - GPS API
  - SMS/Email Notification Service
  - Android/Web Application

→ Communication Interface
  - Wi-Fi
  - Bluetooth

    - HTTPS
    - REST API

5.8 System Features

- Secure User Login
- Real-Time Health Monitoring
- Automatic Fall Detection
- Emergency Alert Generation
- Medication Reminder
- GPS Location Tracking
- Caregiver Dashboard
- Health Report Generation
- Secure Medical Record Storage

6. Assumptions

- Elderly users wear the monitoring device regularly.
- Internet connection is available.
- Wearable sensors provide accurate readings.
- Caregivers and doctors have internet-enabled devices.
- Cloud services are available for data storage.

7. Constraints

- Technical
    - Internet dependency
    - Limited wearable battery life
- Operational
    - Users require basic device knowledge.
- Economic
    - Cost of wearable devices and cloud services.
- Legal
    - Patient data must comply with privacy regulations.
- Time
    - Development must be completed within the project schedule.

8. Requirement Validation

Completeness

All stakeholder requirements have been included, including health monitoring, emergency alerts, medication reminders, and secure data management.

Consistency

The functional and non-functional requirements support the project objectives without conflicts.

Clarity

Requirements are clearly defined and uniquely identified to avoid ambiguity.

Improvements

Future versions may include AI-based disease prediction, voice assistance, smart home integration, and video consultation.

9. Conclusion

The proposed **Elderly Care Monitoring System with Emergency Alerts** provides continuous health monitoring, automatic emergency alerts, medication reminders, and secure health record management. The SRS defines the system requirements clearly and provides a strong foundation for designing and developing a reliable healthcare monitoring solution.